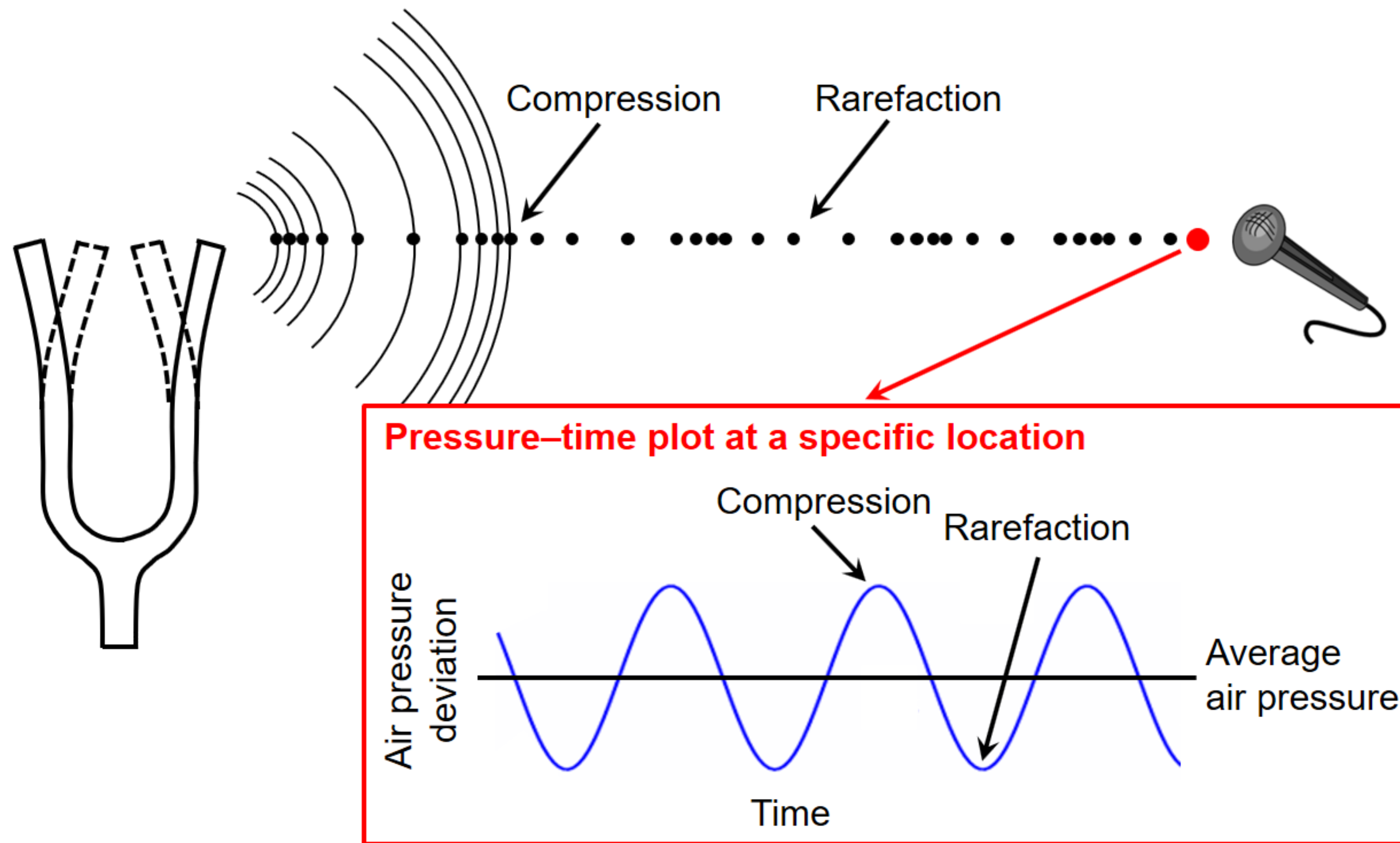


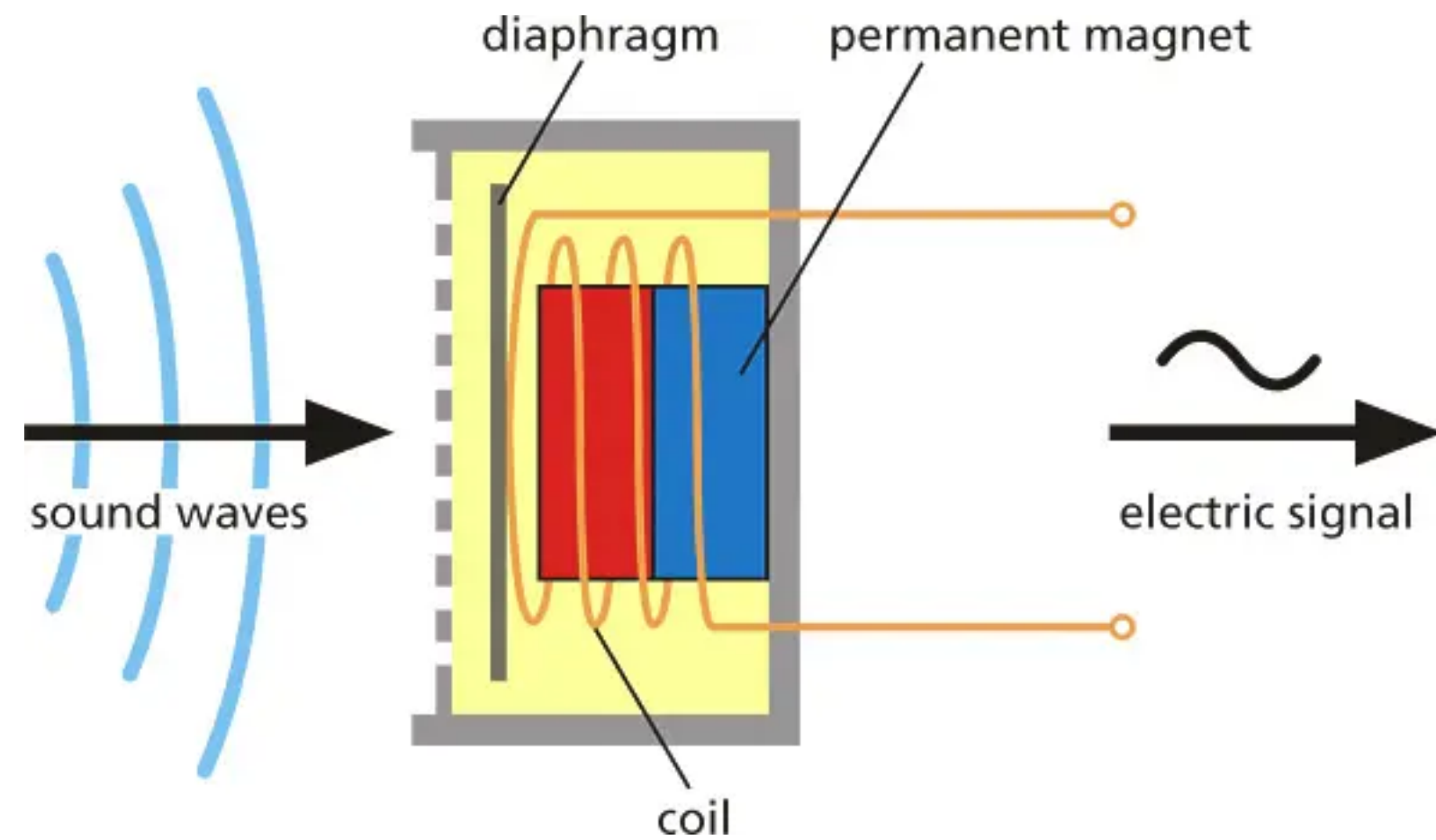
Digital Sound Representation

Sound Recording



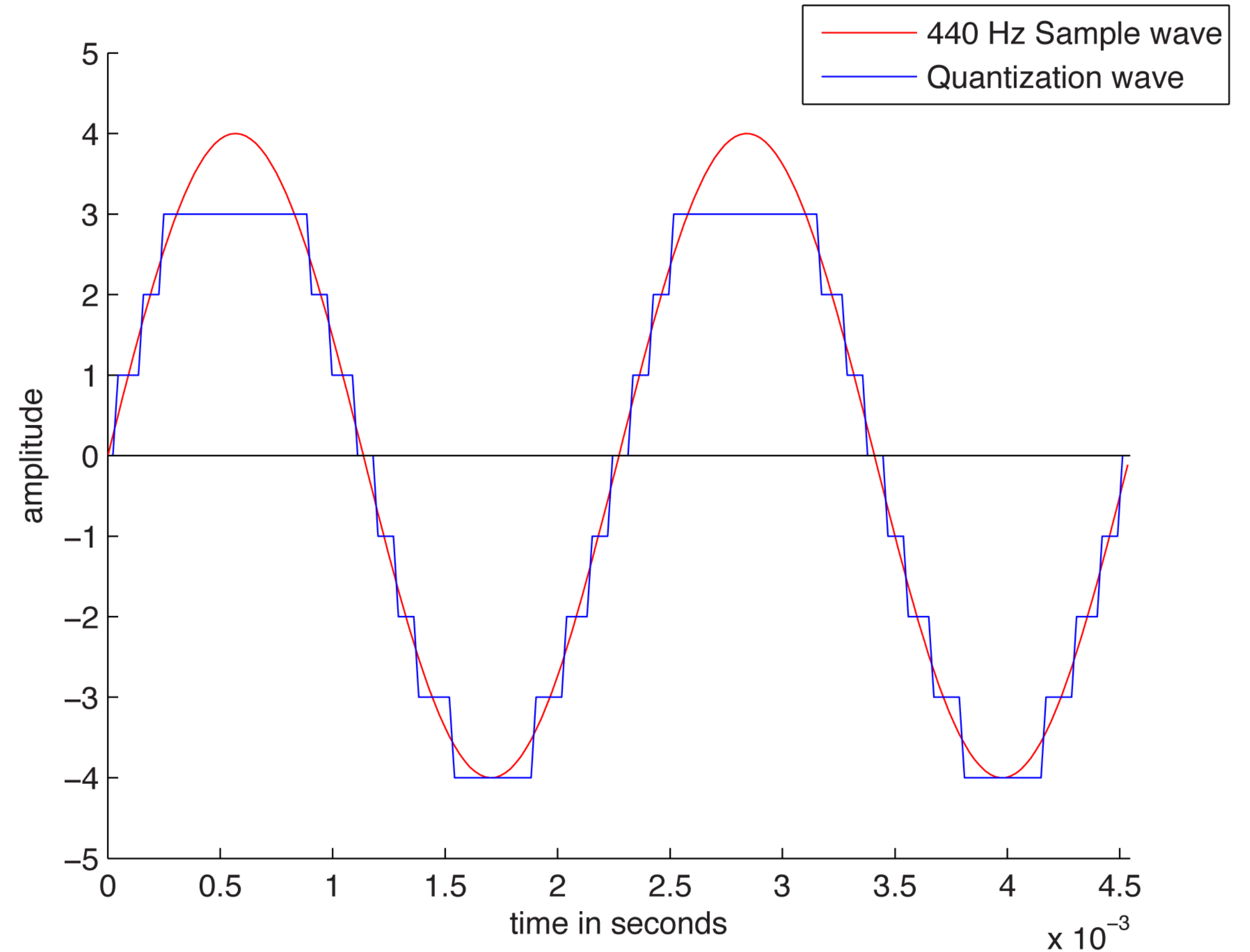
From: Müller, FMP, Springer 2015

Signal Digitisation

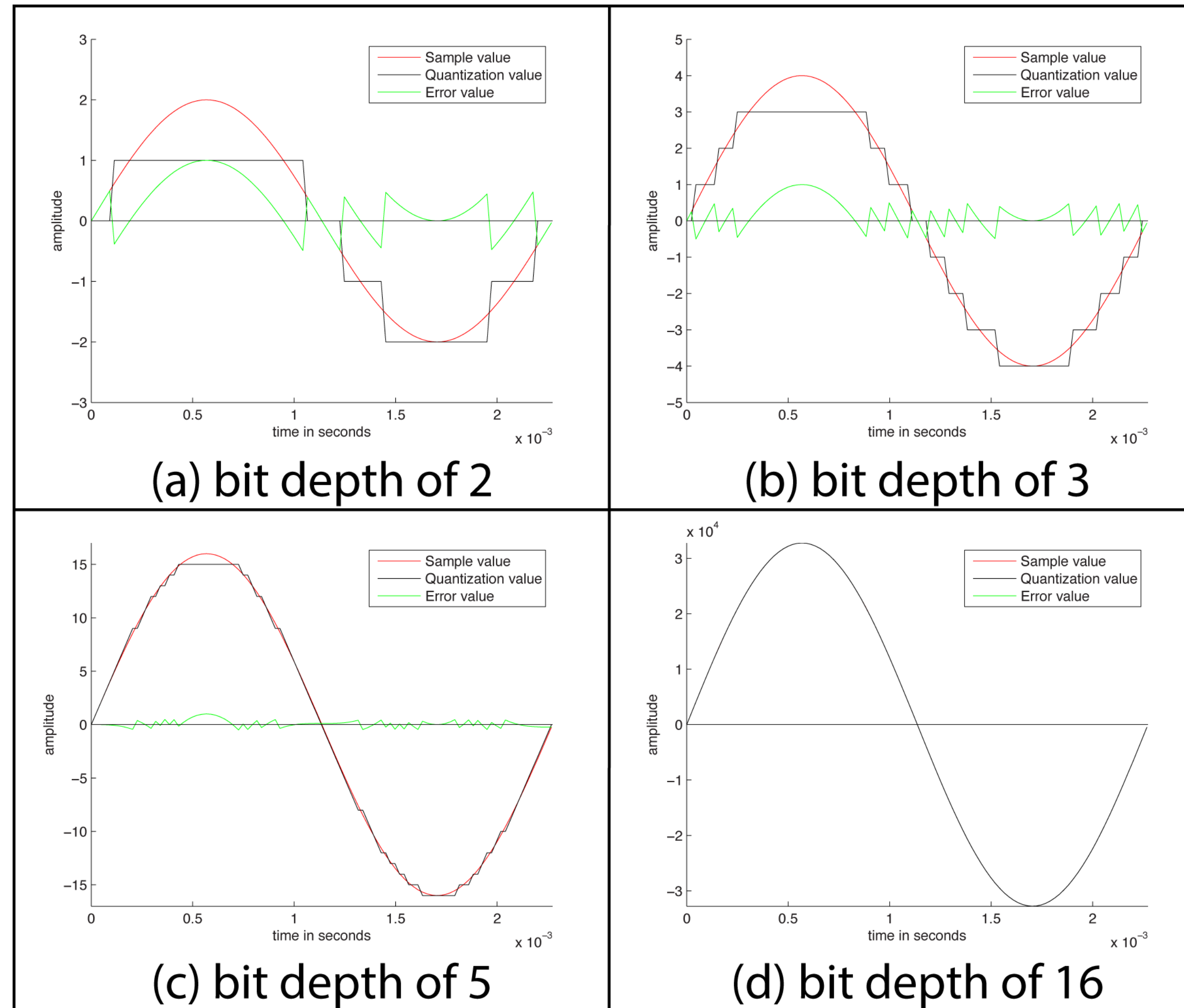


Dynamic Microphone

From: <https://electricalsimplified.blogspot.com/2012/02/dynamic-microphones.html>



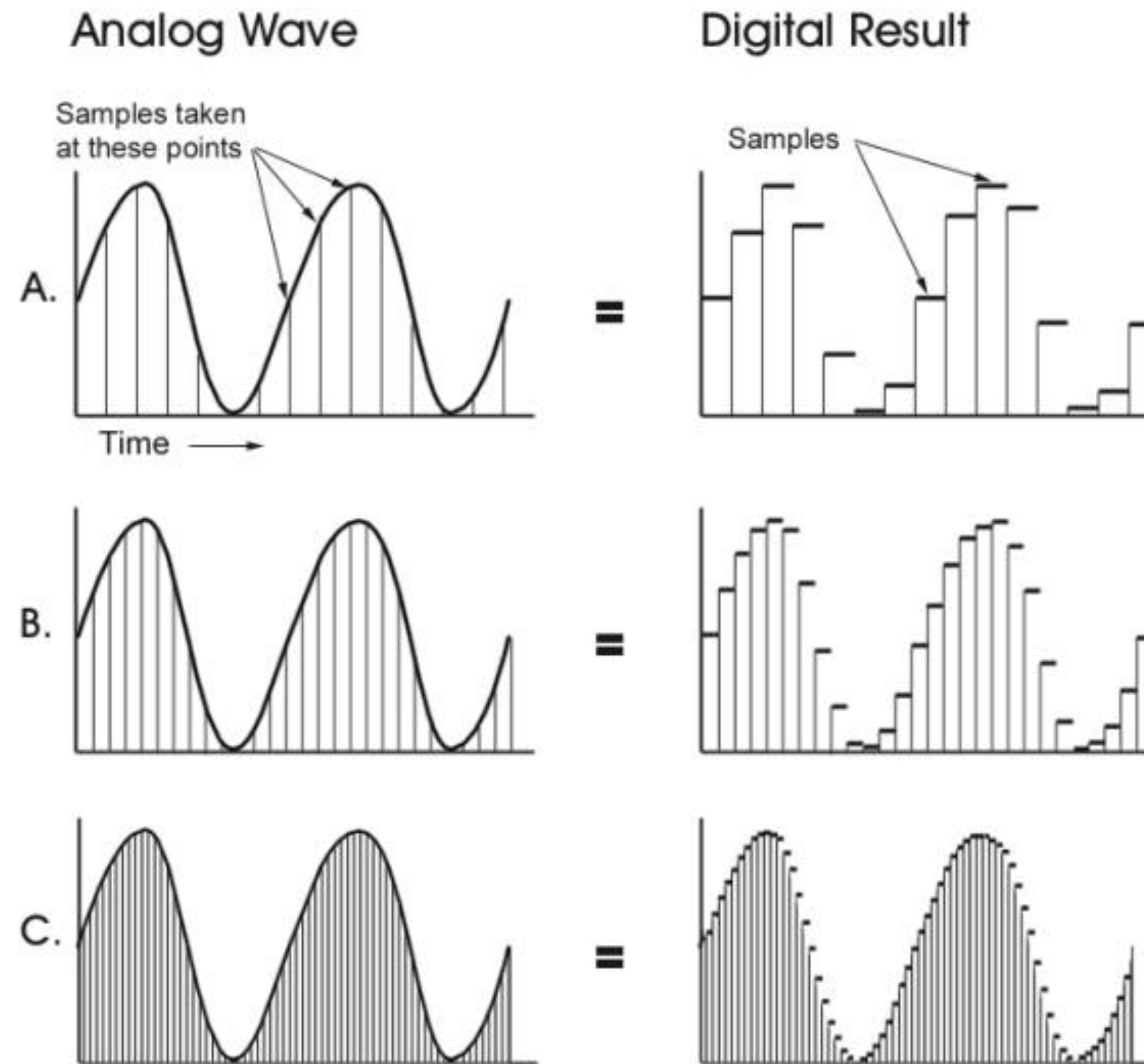
Signal Digitisation - Quantification Rate



From: <https://digitalsoundandmusic.com/5-1-2-digitization/>

Signal Digitisation - Sampling Rate

Increasing Sample Rates

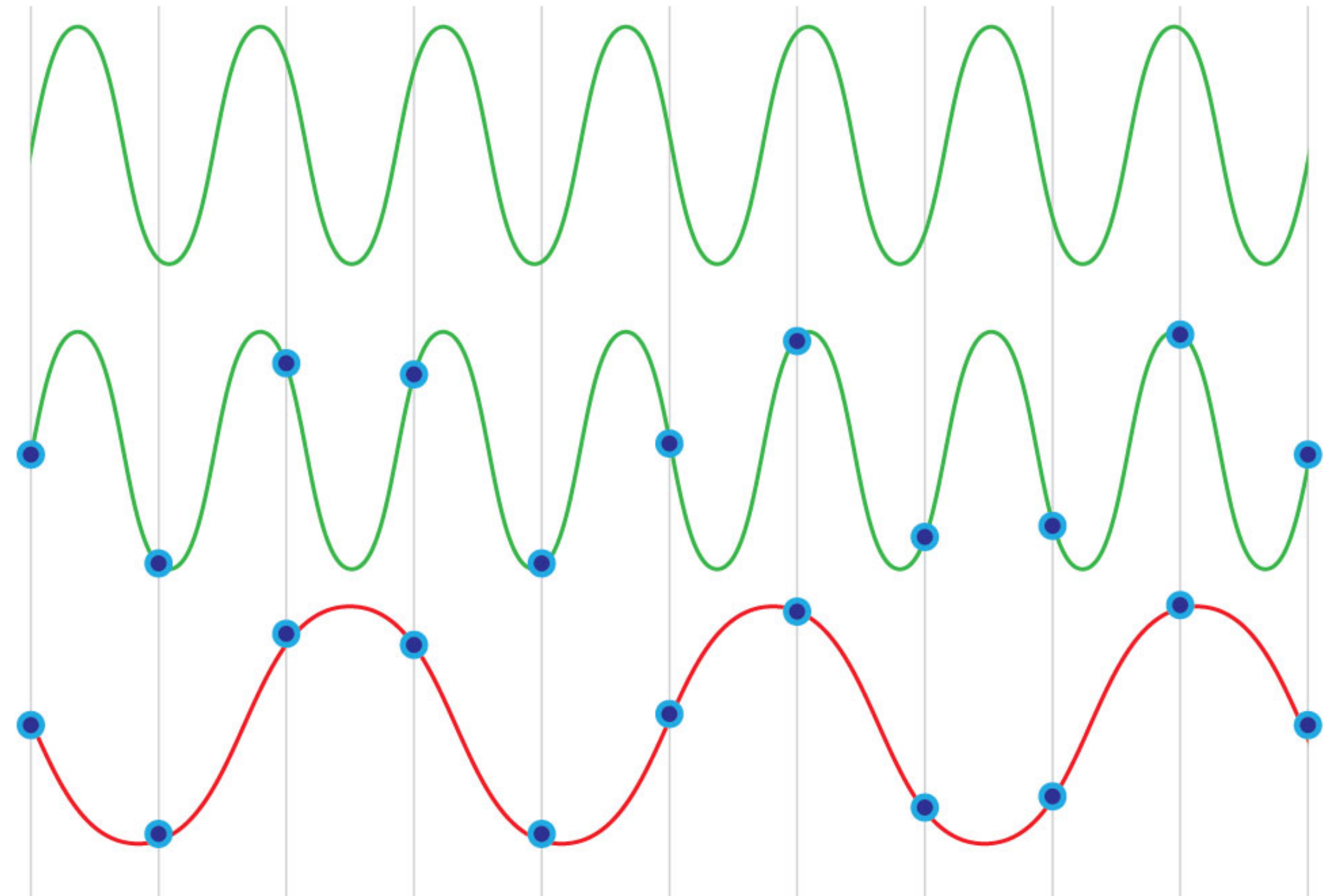


From: <https://www.izotope.com/en/learn/digital-audio-basics-sample-rate-and-bit-depth.html>

Aliasing & Foldover



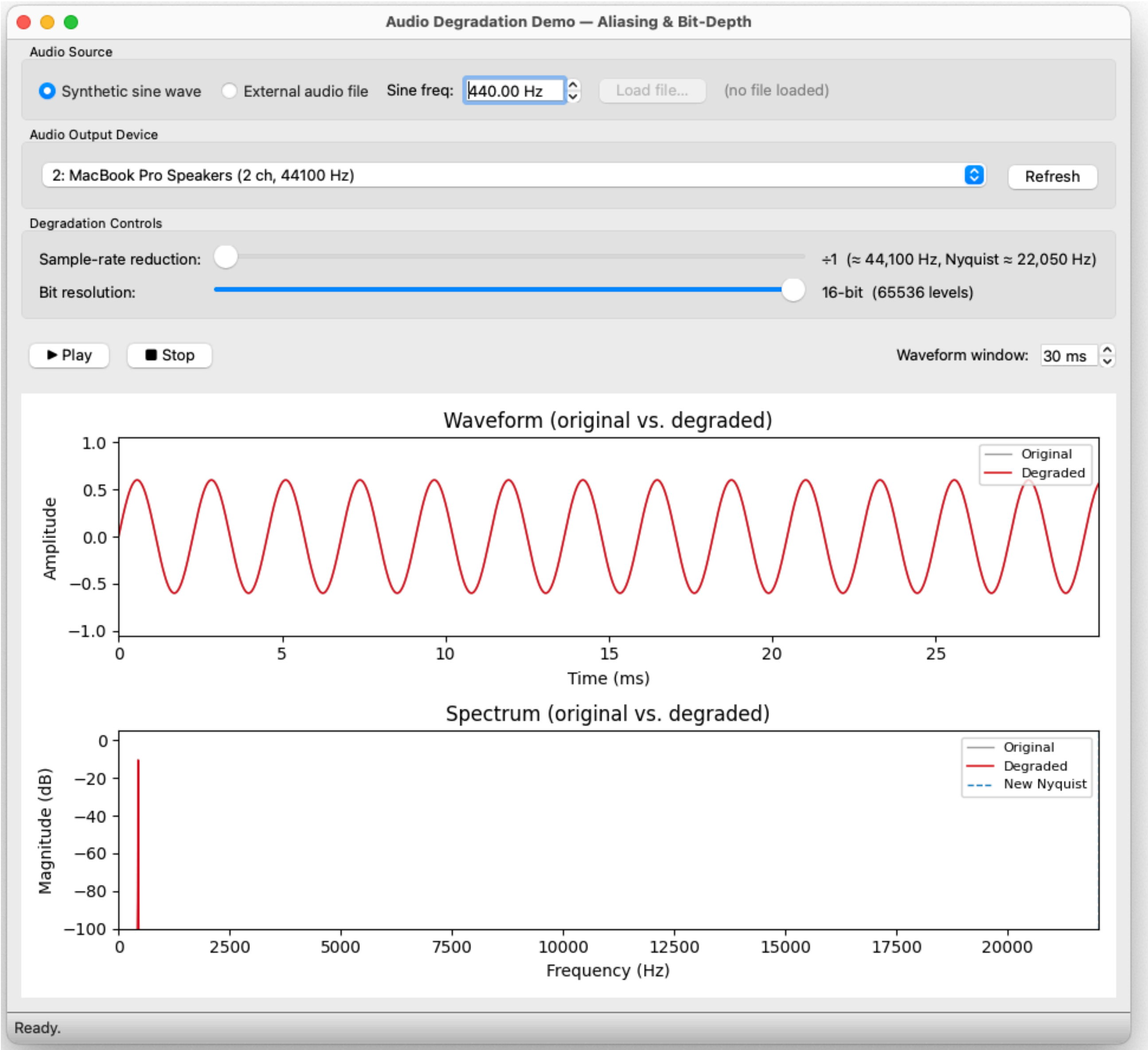
Visual Aliasing



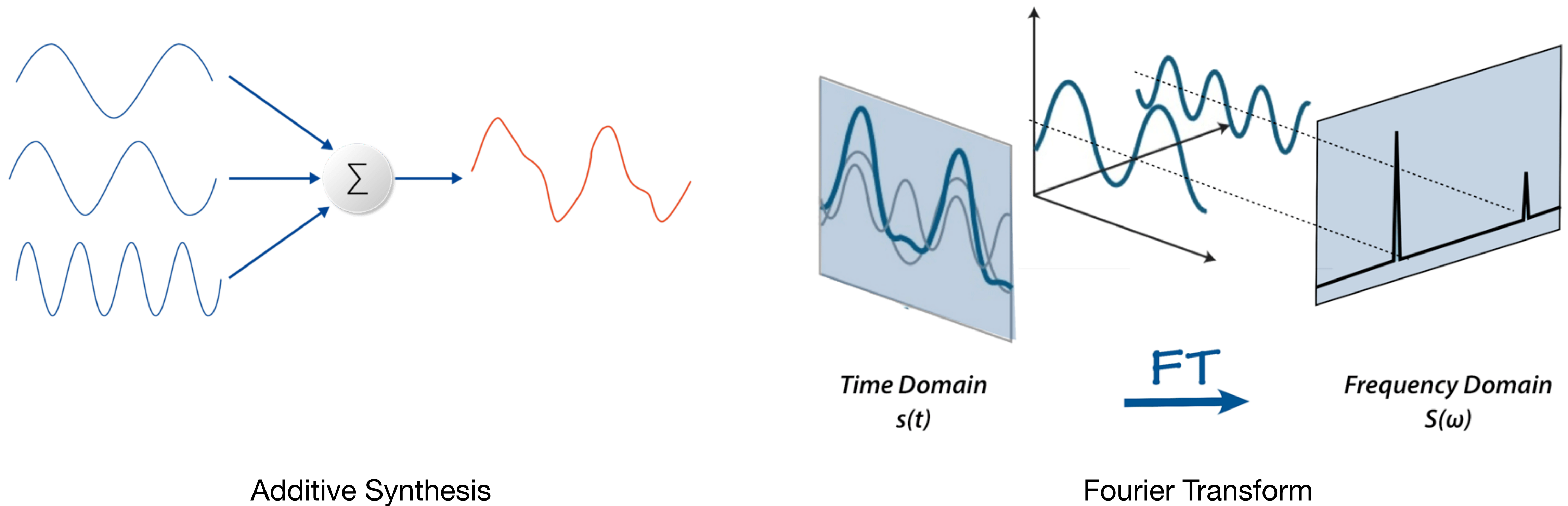
Acoustic Aliasing

From: <https://www.realhd-audio.com/?p=2983>

Audio Degradation Demo

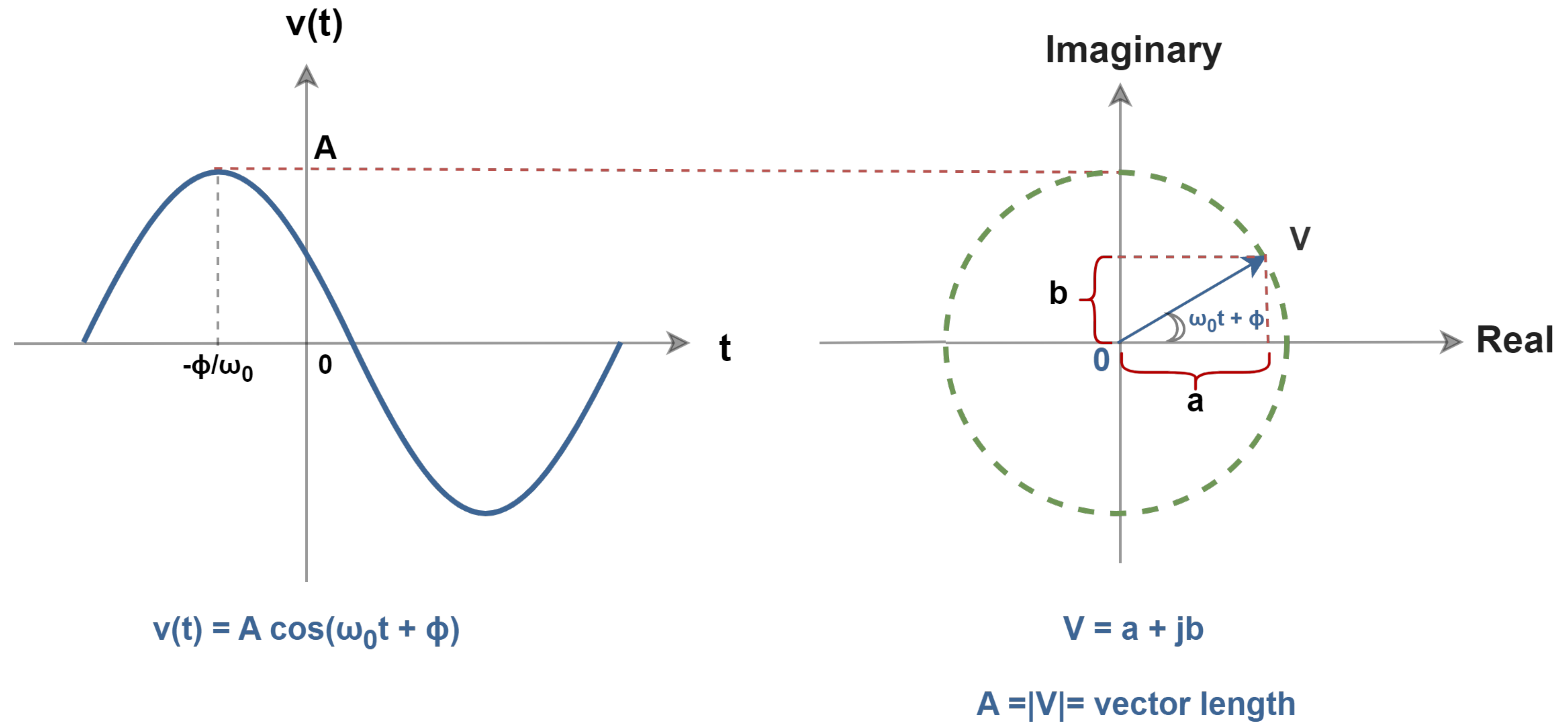


Fourier Transform



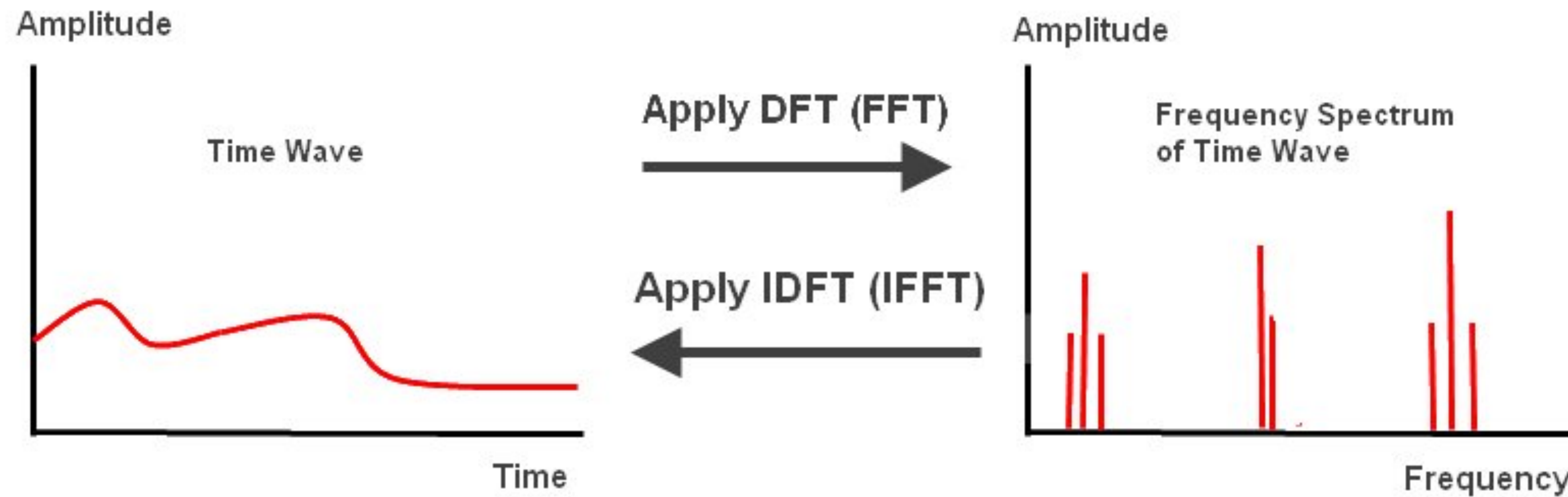
From: <https://medium.com/analytics-vidhya/understanding-the-mel-spectrogram-fca2afa2ce53>

Fourier Transform



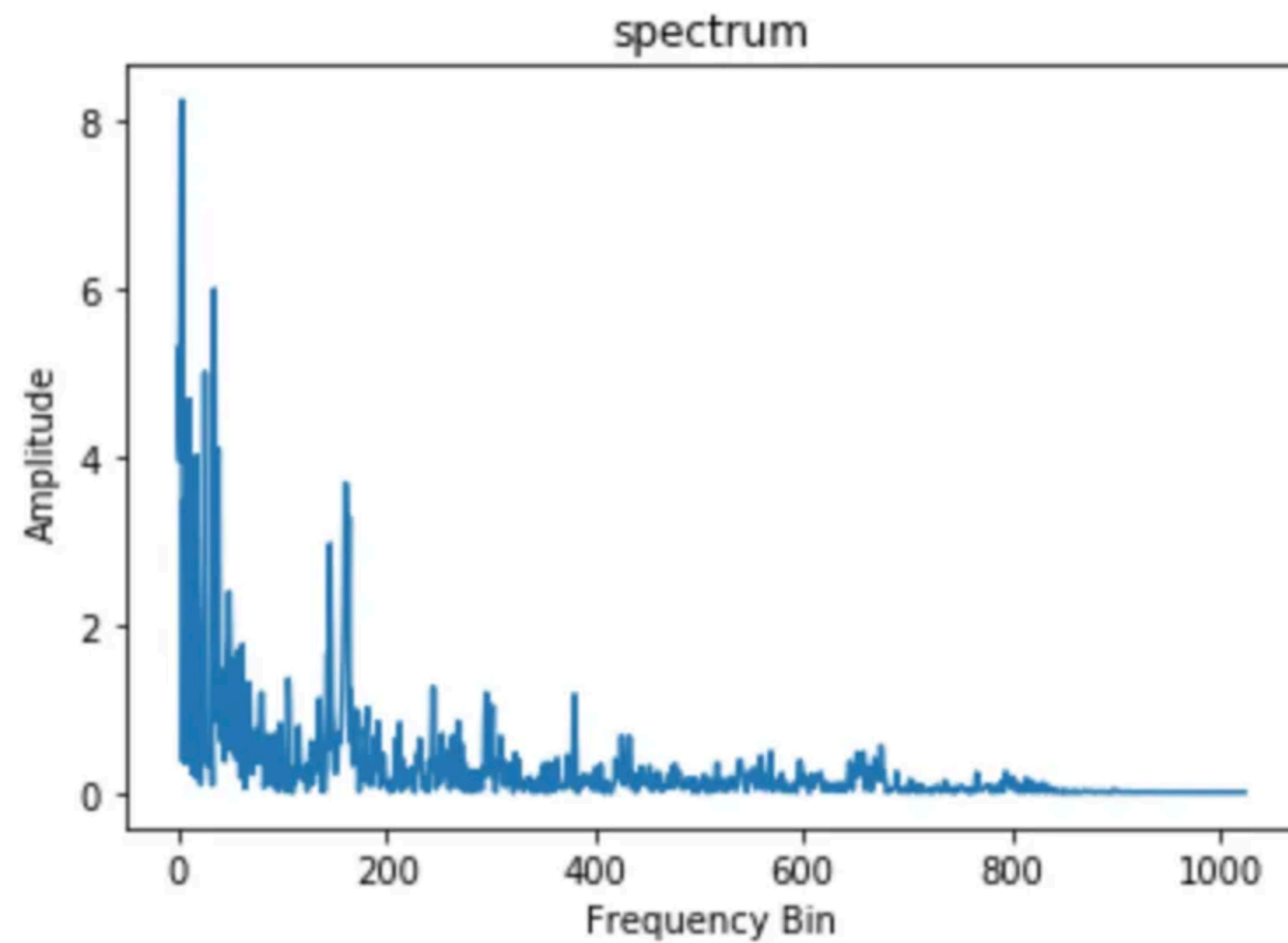
From: <https://www.electronics-lab.com/article/the-fourier-analysis-part-2/>

FFT and IFFT

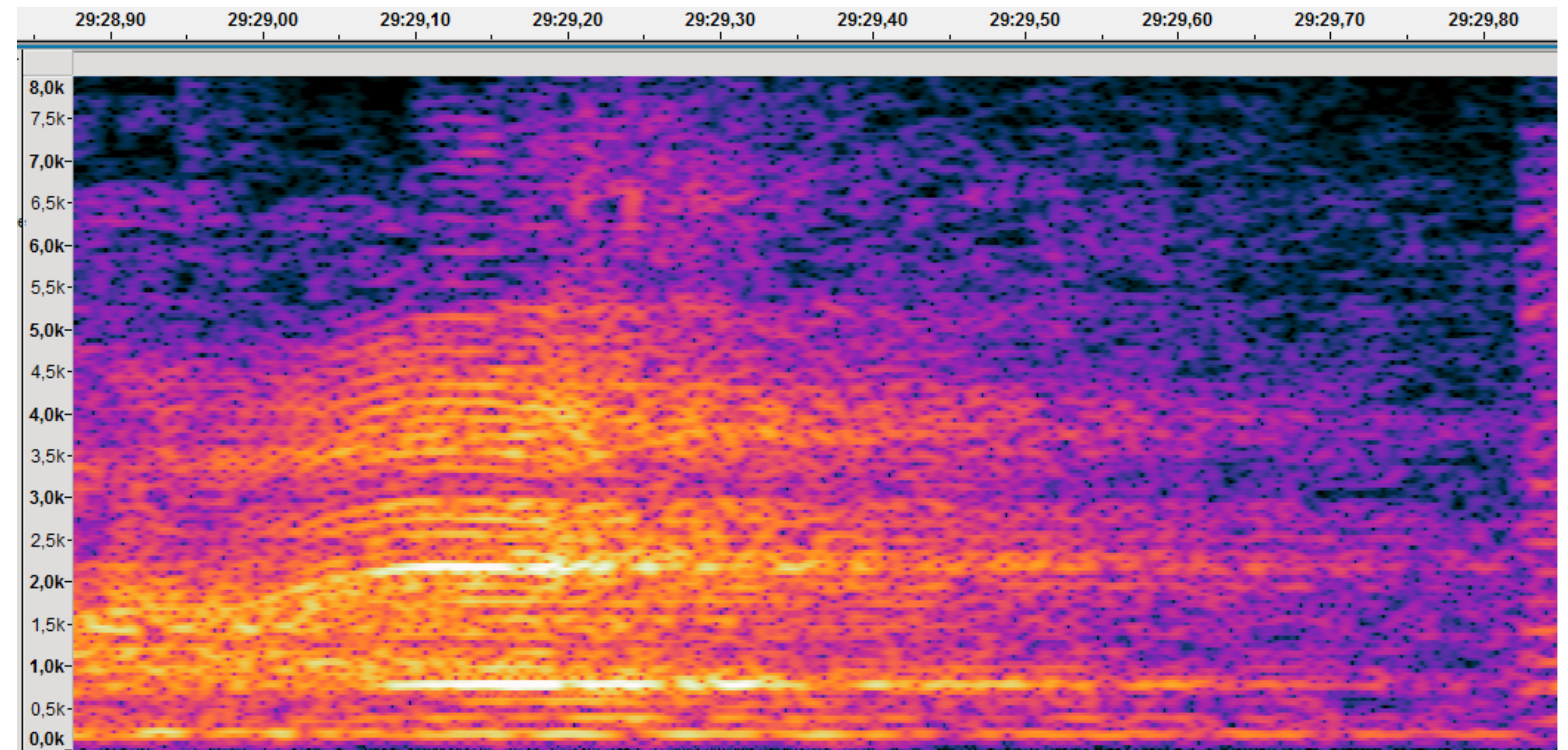


From: https://www.alwayslearn.com/DFT%20and%20FFT%20Tutorial/DFTandFFT_BasicIdea.html

Spectrum and Spectrogram



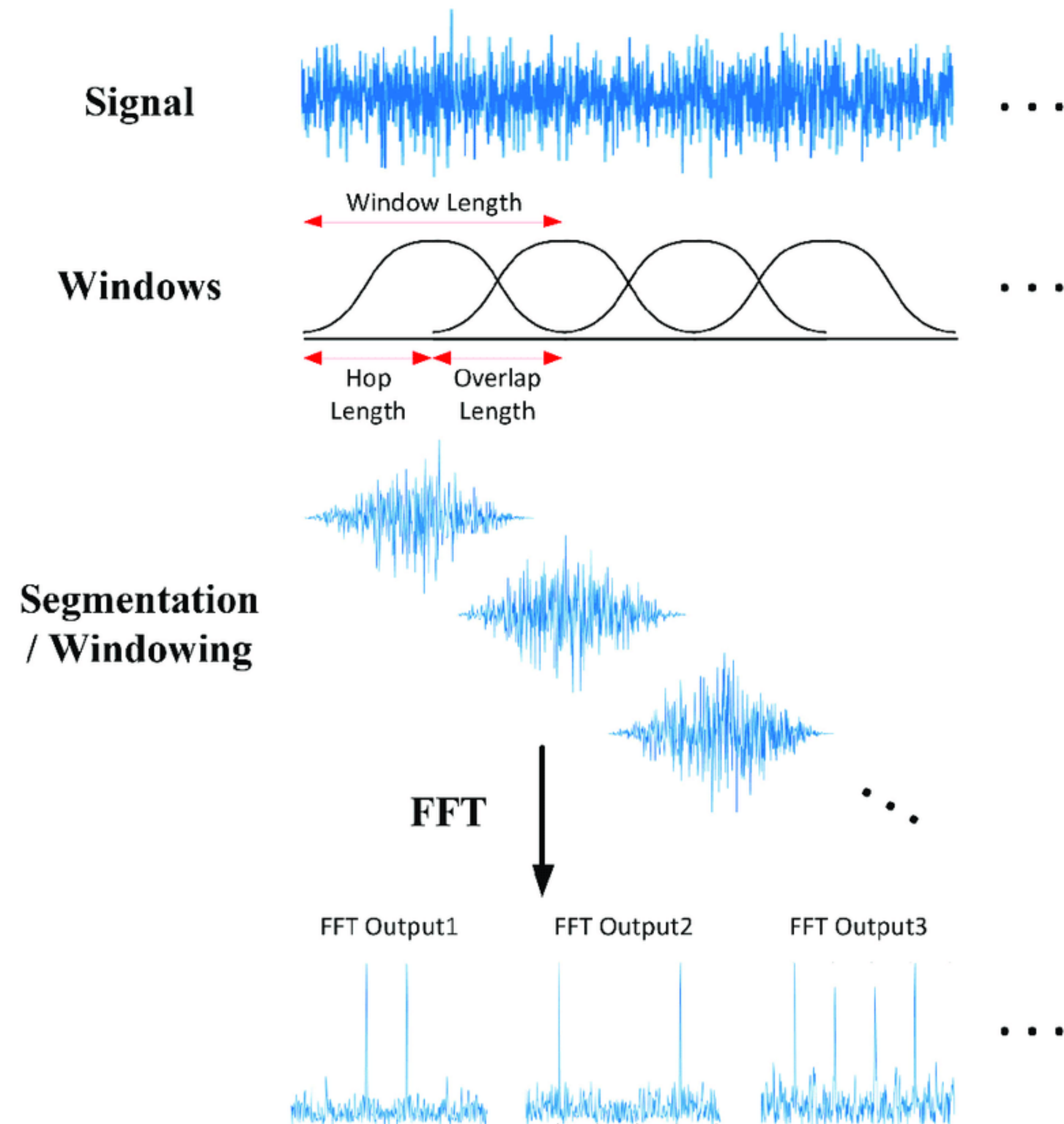
Spectrum



Spectrogram

From: <https://medium.com/analytics-vidhya/understanding-the-mel-spectrogram-fca2afa2ce53>

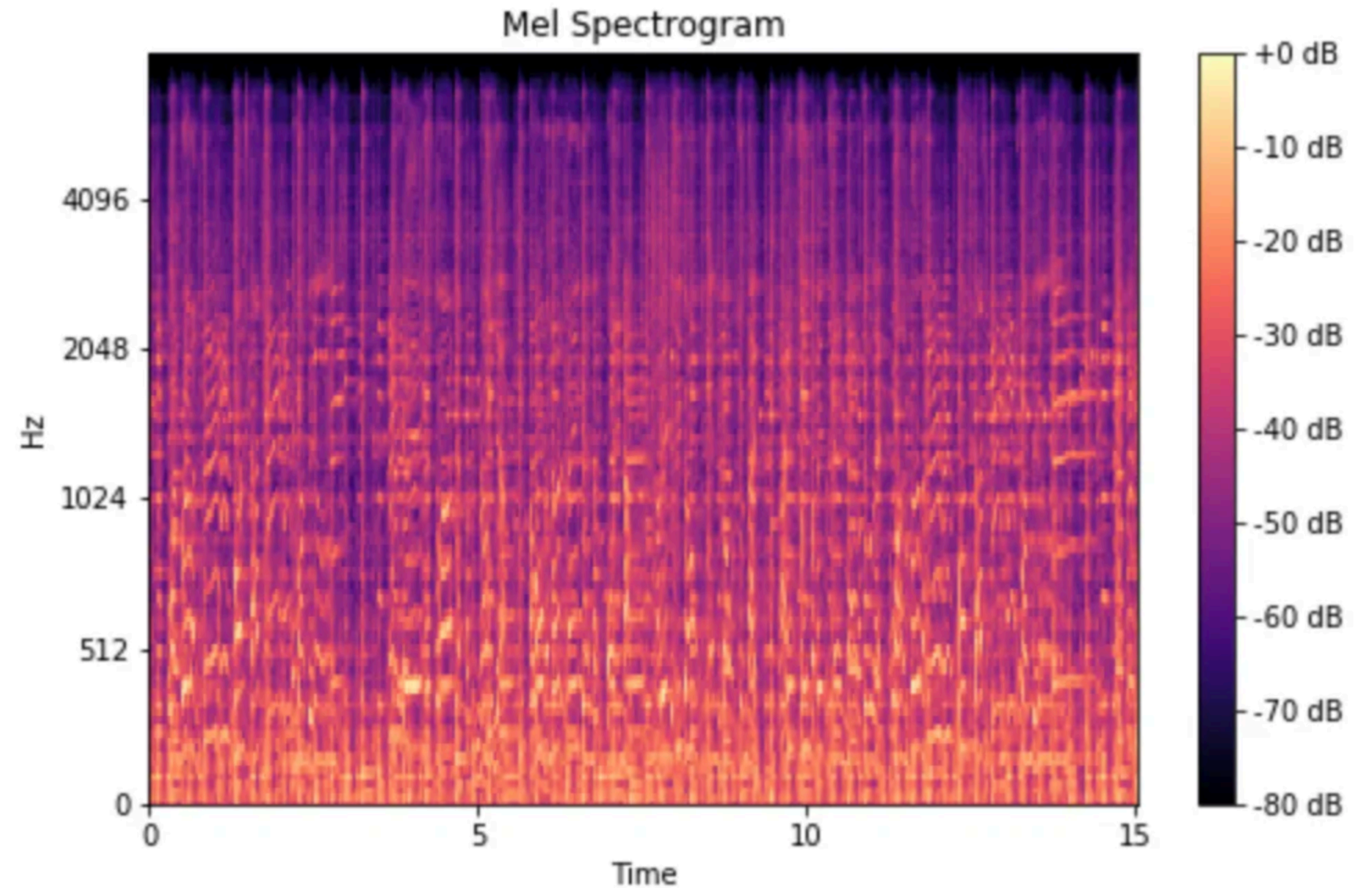
Short-Time Fourier Transform



From: Jeon et al. 2020, Area-Efficient Short-Time Fourier Transform Processor for Time-Frequency Analysis of Non-Stationary Signals

Mel Spectrogram

Mel spectrograms are generated by applying a Fourier transform to a signal and then mapping the resulting frequencies onto the mel scale, a logarithmic scale that is based on human perception of pitch.



From: <https://medium.com/analytics-vidhya/understanding-the-mel-spectrogram-fca2afa2ce53>